
The Asymmetry Principle: A Region-Anchored, Citation-Audited Index of Frontier AI Risk Across Africa

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Abstract

Frontier AI risk on the African continent does not present as a scaled-down version of the profile modeled in the United States, European Union, or China. A field study of 57 interviews with 27 former members of Boko Haram reports use of consumer frontier AI by an armed group in northeast Nigeria (Juelich, 2026); because that evidence is self-reported and not forensically corroborated in public, we treat it as a credible but untriangulated account. TransUnion Africa reports a roughly 1,200% year-over-year rise in deepfake incidents in South Africa, concentrated in banking and fintech. Kenya’s 2022 electoral information environment saw over 550,000 posts screened for toxicity by AI classifiers. We study four anchor states (Egypt, Nigeria, Kenya, South Africa) standing in for North, West, East, and Southern Africa, and add a short unscored section on the Central African mineral supply chain beneath global compute. We build a four-component index (AFAIRI): internet exposure (E), reported scam victimization from one harmonized four-country survey (H), a governance deficit that scores enforcement capacity and not only legislative existence (G), and infrastructure disruption from grid failures and subsea-cable outages (I). A Monte Carlo analysis over index weights and independent input error finds *no* robustly top-ranked state: South Africa has the highest mean and ranks first in about 47% of sampled weightings, Egypt in 31%, Nigeria in 15%, Kenya in 7%, and no state outscores another on every component. An earlier claim that Southern Africa’s top ranking is robust (90–95%) did not survive a construct revision and is withdrawn. Every quantitative claim traces to a cited source, and an appendix logs the corrections made in review, including figures that were fabricated in an earlier draft and removed. We close with a governance architecture scoped to what the regional evidence supports.

1 Introduction: the Asymmetry Principle

Frontier AI safety frameworks built in the United States, European Union, and China assume infrastructure that is not uniform across Africa: stable power grids, formal banking clearinghouses, biological-material screening, and mature statutory enforcement (Anthropic, 2024; European Union, 2024). Where that baseline holds, frontier AI risk is modeled around misalignment, intellectual-property infringement, and long-horizon loss of control. Large parts of the continent instead combine rapid consumer AI adoption with material gaps in that baseline. We call this the **Asymmetry Principle**: the same model capability produces different realized failure modes depending on the infrastructural density and governance capacity of the environment into which it is deployed. This is a qualitative organizing claim, not a fitted function; no panel dataset spanning enough countries and years exists to estimate one, and we do not claim to have built one.

Design. Each region is represented by one anchor state. This is a case-study design, not a regional average, and results should not be read as describing every state in a region. Region stays a useful unit because several phenomena cross borders: Boko Haram’s integration into Islamic State networks connects to a jihadist ecosystem that includes Al-Shabaab in East Africa; West African cyber-fraud actors operate cells across the continent, including in South Africa; and pre-existing governance-capacity gaps now interact with AI-enabled tooling. Central Africa is not scored (Section 6) but is not ignored: Section 5 addresses the Democratic Republic of Congo’s role as the dominant source of mined cobalt, the physical supply-chain dimension of the same asymmetry.

Five claims. The paper is organized around five claims, each tied to specific evidence. **C1 (framing).** The Asymmetry Principle: the same frontier-AI capability produces different realized failure modes across environments, and the four anchor states instantiate four distinct modes rather than one scaled to different sizes. **C2 (headline result).** Under a construct-valid, weight-agnostic Monte Carlo, *no* African anchor state is robustly the highest-risk: no ordering holds across the weight simplex, and no state dominates another on every component (Section 3). **C3 (methodological).** Composite frontier-risk rankings are dominated by construct choice, not data: replacing environmental and inclusion proxies with measures of reported victimization, enforcement, and disruption overturns the earlier draft’s “Southern Africa robust at 90–95%” and drops South Africa’s ranked-first share below 50%. **C4 (evidence integrity).** Under systematic triangulation, the strongest *corroborated* evidence of African frontier-AI misuse is fraud, not terrorism: Nigeria-origin AI-assisted scam networks are platform-corroborated (OpenAI, 2025), whereas the widely cited insurgent-AI account is self-reported and untriangulated. **C5 (governance).** Enforcement, not legislation, is the binding governance gap: scoring enforcement capacity separately from statute reveals that a state with a published AI strategy (Egypt) can carry the largest enforcement gap while the weakest strategy status (Nigeria) coexists with the strongest penalty record. We also release a citation-and-methodology audit log that documents, rather than hides, what earlier drafts got wrong, including figures that were fabricated and removed.

2 Four anchor states

North Africa (Egypt). Strong laboratory biosafety practice assessed through the WHO Joint External Evaluation, alongside the largest governance-enforcement gap in the set: its data-protection regime begins enforcement only in October 2026. Egypt also ran national load-shedding for roughly 202 days in 2024. **West Africa (Nigeria).** The self-reported insurgent-misuse account (Juelich, 2026) and the most severe measured infrastructure disruption (12 national grid collapses in 2024), coupled with, on a penalty-issuance proxy, the strongest data-protection enforcement record (a USD 32.8M action against Meta). **East Africa (Kenya).** A mobile-first ecosystem, the highest reported scam victimization of the four states, electoral information-integrity strain in 2022, and repeated 2024 blackouts. **Southern Africa (South Africa).** A roughly 1,200% year-over-year rise in deep-fake incidents concentrated in banking and fintech (TransUnion Africa, 2025), the highest internet exposure, and the highest mean index score.

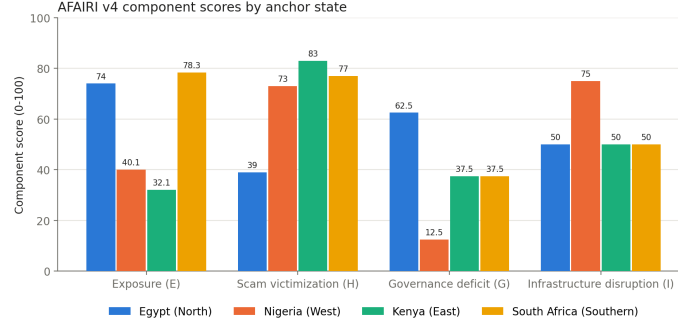
3 The AFAIRI composite index

AFAIRI is a composite proxy index in the family of the Human Development Index or ND-GAIN: a few individually cited indicators combined under a disclosed weighting scheme. It is *not* validated against an external ground truth of realized AI-enabled harm, because no such dataset exists across these states. It is a structured comparison of four documented case studies, not a risk-prediction model.

Specification. $AFAIRI(s) = w_E E + w_H H + w_G G + w_I I$ with $w_E + w_H + w_G + w_I = 1$. *Exposure* E is 2023 internet penetration (World Bank WDI, untransformed). *Reported scam victimization* H is the share of adults reporting a scam in the previous 12 months, from the Global Anti-Scam Alliance’s *State of Scams in Africa 2025*, a survey of 4,000 adults covering exactly these four states. *Governance deficit* $G = 100 - (A + B + C)$ scores AI-strategy status (A), AI-specificity of cybercrime law (B), and enforcement capacity (C , keyed to whether the data-protection authority has issued monetary penalties and of what size). *Infrastructure disruption* I sums a grid-failure tier and a subsea-cable-outage score for calendar year 2024. Every threshold (for example USD 1M,

Table 1: AFAIRI v4 component matrix (0–100 scale). See Figure 1.

Anchor state (region)	<i>E</i> exposure	<i>H</i> victimization	<i>G</i> gov. deficit	<i>I</i> disruption
Egypt (North)	74.0	39.0	62.5	50.0
Nigeria (West)	40.1	73.0	12.5	75.0
Kenya (East)	32.1	83.0	37.5	50.0
South Africa (Southern)	78.3	77.0	37.5	50.0



Sources: World Bank WDI 2023 (E); GASA State of Scams in Africa 2025 (H); rubric in Section 6.3 (G, I).

Figure 1: AFAIRI v4 component scores by anchor state (*E*, *H*, *G*, *I*; 0–100), values printed on each bar. Data as in Table 1.

90 days, six collapses) was fixed before the scores were computed. Full rubrics, source values, and per-criterion limitations are in the manuscript; Table 1 gives the resulting component matrix.

Sensitivity analysis. We drew $N = 10^4$ weight vectors from a symmetric Dirichlet(1, 1, 1, 1) (uniform over the weight simplex), recomputed AFAIRI for each state, and repeated with independent multiplicative perturbations of each of the 16 inputs at $\pm 10\%$ and $\pm 25\%$, and with a coupling term restored as a robustness run (seed 42). “Share ranked first” is the fraction of the sampled weight space in which a state scores highest; it is *not* the probability that a state is in fact the riskiest, because the uniform prior over weights is itself a modeling choice. Reported ranges are 2.5th–97.5th percentiles of the score across weightings: sensitivity ranges, not confidence intervals.

What the index supports. South Africa has the highest mean, ranks first most often, and outscores each other state in a majority of weightings (66% vs. Egypt, 83% vs. Nigeria, 87% vs. Kenya). The result is stable under input error and the coupling run (Table 2; ranked-first shares move by at most a few points). What is *not* supported is a claim that any state is robustly highest-risk: South Africa ranks first in fewer than half of weightings and Egypt in almost a third, because the two are high on different components (South Africa on victimization, Egypt on governance deficit), and the Kenya–Nigeria ordering is effectively a coin flip. The earlier “Southern Africa robust at 90–95%” headline came from constructs that external review found invalid and is withdrawn.

Verification. A separate script recomputed every component by hand from the rubric (all matched); confirmed that each state’s Dirichlet mean equals the simple average of its four components (matched to within 0.07 on a fresh 200,000-draw simulation); and recomputed the ranked-first shares exactly on a $\approx 176,000$ -point grid over the simplex (South Africa 46.2%, Egypt 31.1%, Nigeria 15.3%, Kenya 7.3%), agreeing with the random sampling to within one point. Re-running the previous draft’s code is how a jitter implementation error (a single random factor applied to all four components, rather than independent perturbation) was found and fixed.

4 Threat-actor characterization

We label each actor by an evidence tier rather than asserting uniform confidence. The earlier label “PRIMARY-VERIFIED” for the Boko Haram study conflated reading a source with verifying its claims and is retired. **Boko Haram / ISWAP** (West Africa): reported use of ChatGPT, Claude,

Table 2: Weight-only sensitivity ($N = 10^4$ Dirichlet draws). No robustly top-ranked state.

State	mean	range (2.5–97.5%)	ranked first
South Africa	60.8	45.3–74.5	47.1%
Egypt	56.4	45.0–67.7	31.0%
Nigeria	50.2	25.8–69.6	15.2%
Kenya	50.6	37.2–70.8	6.7%

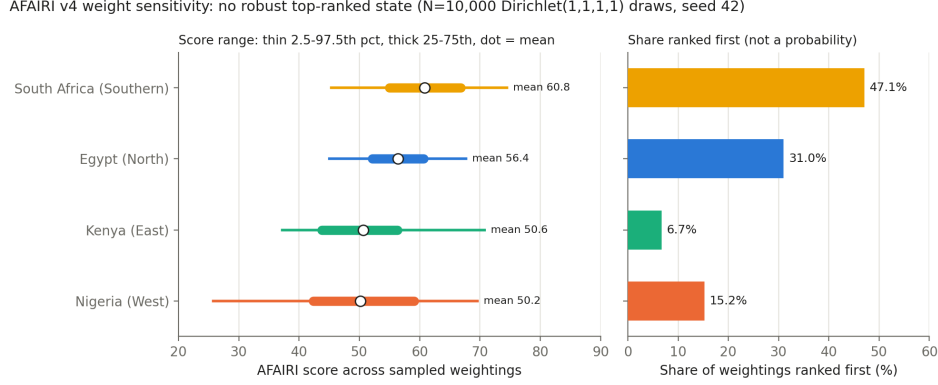


Figure 2: Weight sensitivity ($N = 10^4$ Dirichlet(1, 1, 1, 1) draws, seed 42). Left: distribution of each state’s score across weightings (thin line 2.5–97.5%, thick line 25–75%, dot = mean). Right: share of weightings in which each state ranks first. Sensitivity summaries, not confidence intervals or probabilities.

117 Gemini, Grok, Meta AI, and DeepSeek for tactical planning, explosives, weapons servicing, and
 118 operational security, with unlinkable accounts and pretextual framing to defeat content restrictions
 119 (Juelich, 2026). Tier: *primary-read, self-reported, untriangulated*; a May 2026 seizure of 400+
 120 Starlink terminals corroborates connectivity but not AI use, and no seized-device forensics or plat-
 121 form disclosure attributing activity to these groups was located. **AI-Shabaab** (East Africa): AI-
 122 assisted propaganda translation, attributed to a 2025 UN monitoring-team report via secondary cov-
 123 erage (*secondary-corroborated*; the primary UN document was not read). **“Yahoo Boys” / “AI**
 124 **Boys”** (West Africa, global targets): AI-generated image, voice, and message content for fraud;
 125 *platform-corroborated* at the level of Nigeria-origin AI-assisted scam networks, via OpenAI’s Oc-
 126 tober 2025 threat-intelligence disclosure. **Black Axe** (transnational): a structured syndicate with
 127 law-enforcement-confirmed fraud (INTERPOL Operation Jackal III; Europol arrests) but *no* con-
 128 firmed Black Axe-specific AI use, and none is claimed. Threat-actor activity is deliberately *not* an
 129 index input.

130 5 Governance architecture and the compute supply chain

131 Each pillar is scoped to its evidence. **Pillar 1: AU-level red-teaming and evaluation capac-**
 132 **ity**, motivated by the self-reported workaround pattern (unlinkable accounts, pretextual framing),
 133 which is cheap to add to evaluation suites whether or not the interview claims of uplift hold, plus a
 134 forensic-corroboration workstream so future threat claims can be checked against account and device
 135 evidence. **Pillar 2: additionality and base-load isolation for new compute**, an environmental-
 136 governance recommendation kept separate from the safety-risk index. **Pillar 3: application-layer**
 137 **provenance with funded enforcement**, motivated by South Africa’s deepfake and digital-banking
 138 fraud and platform-corroborated Nigeria-origin fraud: a C2PA-style provenance standard is only
 139 as effective as the authority enforcing it, so Pillar 3 pairs provenance rules with funded data- and
 140 consumer-protection enforcement and algorithmic recourse for victims.

141 **The unscored supply chain.** The asymmetry also runs through the physical supply chain be-
 142 neath compute. The DRC accounted for about 74% of world mined cobalt in 2023, most of it from
 143 Chinese-owned or -controlled operations (USGS, 2024). Cobalt is used in the lithium-ion backup

144 and UPS systems that data centers and telecom towers depend on. The states whose infrastructure
145 resilience we score therefore depend, like the rest of the world, on a mineral supply chain concen-
146 trated in a Central African state whose governance and conflict conditions fall outside the index. We
147 flag this as priority future work rather than fill it with unverified figures.

148 6 Limitations

149 Four anchor states stand in for four regions; results do not generalize to every state in a region, and
150 Central Africa is only discussed qualitatively. *H* relies on one industry-alliance survey whose coun-
151 try figures were taken from its public summary and independent coverage, not the members-only
152 full report, and it disagrees with national loss statistics on South Africa's relative position. Gover-
153 nance and infrastructure thresholds are author-chosen, not empirically derived. Penalty issuance is
154 an imperfect proxy for enforcement capacity; oversight budgets were not available in comparable
155 form. Section 4's Boko Haram evidence rests on self-report and remains untriangulated, and the
156 Al-Shabaab claim rests on secondary reporting of an unread UN document. The Egyptian biosecu-
157 rity JEE dates from 2018. Above all, AFAIRI is not validated against realized AI-enabled harm; an
158 earlier pilot correlating index scores with a labeled harm-event set in this project's repository was
159 underpowered and found no significant association.

160 7 Conclusion

161 The four anchor states show structurally different ways in which frontier AI capability meets variable
162 infrastructure and enforcement capacity: self-reported insurgent misuse and severe disruption in
163 Nigeria, measured digital-fraud harm in South Africa, electoral information strain and repeated grid
164 failures in Kenya, and an enforcement gap in Egypt alongside strong laboratory biosafety practice.
165 Revising the index in response to review changed its headline: once environmental and inclusion
166 proxies are replaced with measures of reported victimization, disruption, and enforcement, South
167 Africa remains the most frequently top-ranked state but not a robust one, and Egypt's unenforced
168 data-protection regime makes it a close second. That is a less striking conclusion than an earlier
169 draft's, and a more defensible one. Grounding African AI-risk analysis in cited, regionally specific
170 evidence, and being explicit about what the evidence does not support, is the contribution we most
171 want to carry from roots to reach.

172 Use of AI assistance

173 AI tools (a large language model assistant) were used for drafting and editing prose, for organiz-
174 ing the citation-audit workflow, and for writing the index computation and verification code. Ev-
175 ery quantitative claim traces to a cited primary or authoritative secondary source that the authors
176 checked, and the authors are responsible for the content and conclusions.

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